

Permutation groups — homework problems

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Submission instructions. To submit homework problems, create an Overleaf project with the title `year_your_name_pg`. Then provide me the editing rights to the project. To pass the course it is required to solve **all problems** in this document **by the end of the academic year**.

Problem 1. Recall that a simple graph is a tuple (V, E) , where V is a finite set and $E \subseteq \binom{V}{2}$. A digraph is a tuple (V, E) , where $E \subseteq V^2$.

- (1) Construct a simple graph X such that $\text{Aut}(X)$ is isomorphic to the cyclic group $\mathbb{Z}_n$.
- (2) Let $G = \{g_1, \dots, g_n\}$ be a group. Define a colored digraph X with $V(X) = G$ such that there is a directed edge of color k from g_i to g_j if

$$g_i g_j^{-1} = g_k \iff g_i = g_k g_j.$$

What is $\text{Aut}(X)$? Note that an automorphism of a colored digraph has to preserve the colors and also the orientations of the edges.

- (3) (Frucht's theorem) Given a finite group G , there exists a simple graph X with $\text{Aut}(X) \cong G$.

Problem 2. A graph is *asymmetric* if its automorphism group is trivial. Let X be a graph with the vertex set V of size n . The set of all graphs isomorphic to X is called the *isomorphism class* of X . The isomorphism classes partition the set of all graphs with vertex set V .

- (1) Construct an asymmetric graph.
- (2) Prove that the size of the isomorphism class containing X is

$$\frac{n!}{|\text{Aut}(X)|}.$$

Strategy: Interpret the isomorphism classes as orbits of a suitable action of $\text{Sym}(V)$. Then apply the orbit-stabilizer theorem.

- (3) Prove that the number of isomorphism classes of graphs on n vertices is at most

$$(1 + o(1)) \frac{2^{\binom{n}{2}}}{n!}.$$

Recall that $o(1)$ is a shorthand for a function that tends to 0 as $n \rightarrow \infty$.

Strategy: The main tool is the Burnside's lemma.¹ The *support* of a permutation is the number of points it does not fix.

- (a) For $g \in \text{Sym}(V)$, let $\text{orb}_2(g)$ be the number of orbits of g in its induced action on $\binom{V}{2}$. Show that the number of isomorphism classes of graphs on V is equal to

$$\frac{1}{n!} \sum_{g \in \text{Sym}(V)} 2^{\text{orb}_2(g)}.$$

- (b) Suppose that $g \in \text{Sym}(V)$ has support $2r$. Show that the maximum value of $\text{orb}_2(g)$ is realized by a permutation with exactly r cycles of length 2.
- (c) To estimate the sum from (a), we partition the permutations in $\text{Sym}(V)$ into classes $\mathcal{C}_1$, $\mathcal{C}_2$ and $\mathcal{C}_3$. Fix an even integer $m \leq n - 2$. The class $\mathcal{C}_1 = \{1\}$, $\mathcal{C}_2$ contains permutations with support at most m , and $\mathcal{C}_3$ contains the remaining permutations. Find what types of permutations in $\mathcal{C}_2$ and $\mathcal{C}_3$ give the maximum values of $\text{orb}_2(g)$ and use this to bound the sum in (a).
- (4) (Almost all graphs are asymmetric) Let $\mu(n)$ be the proportion of isomorphism classes of asymmetric graphs on V , where $|V| = n$. Argue that $\mu(n) \rightarrow 1$ as $n \rightarrow \infty$. Deduce the same about the proportion of asymmetric graphs.

¹also known as the Cauchy–Frobenius lemma, or the orbit-counting theorem